AFS–0576

Sub. Code
7MCH1C1

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Chemistry

ORGANIC CHEMISTRY –I

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Explain why acetylene $(pKa = 25)$ is more acidic than ethylene $(pKa = 36)$, which in turn is more acidic than ethane $(pKa = 42)$.

2. What is meant by sacrificial hyper conjugation? Give example.

3. Explain the terms optical activity and optical isomerism.

4. Define fischer projection formula, write the correct form of fischer projection for erythro $CH_3CHBrCH.Cl.CHO$.

5. Cyclo propene is not aromatic. while cyclopropenyl cation is aromatic-why?

6. Although there are eigth free electrons, phenylanion is aromatic- Explain.

7. Arrange the following carbanions in order of increasing stability. $Ph\overline{C}H_2, \overline{C}H_2NO_2, \overline{C}H_2CO_2\varepsilon t, \overline{C}H_2COCH_3$.

8. Comment : nitrobenzene can be used as a solvent in friedel-crafts reaction.

9. Comment : Allyl bromide undergoes S_N2 reactions slightly more rapidly than does ethyl bromide.

10. What are ambident nucleophiles and substrates? Give examples.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) With suitable example explain how steric effect influences the acidity and basicity of a molecule.

Or

(b) Discuss the quantitative treatment of the effect of structure on reactivity.

12. (a) Discuss the stereochemistry in allenes and spirane compounds.

Or

(b) Give a detailed account on the optical isomerism exhibited by nitrogen componds.

13. (a) Discuss the homoaromaticity in 2,4,8 and 10 π electron systems.

Or

(b) Give an account on alternant and non-alternant hydrocarbons.

14. (a) Discuss about the stability of free radicals and carbocations.

Or

(b) (i) Methylation of toluene at $0°C$ gives a mixture of O-and p-xylene, while at $80°C$ it gives mainly m-xylene. Explain (2.5)

(ii) state and explain Hammond postulate. (2.5)

15. (a) Write notes on nucleophilic substitution at allylic and vinylic carbon.

Or

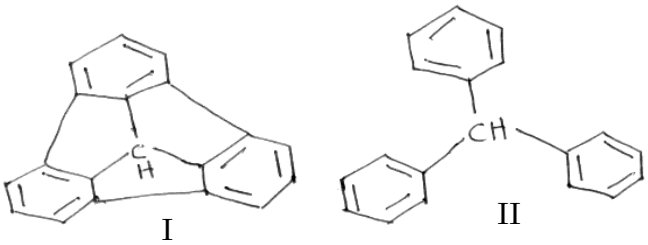
(b) Discuss the mechanism and stereochemistry involved in E_1 reaction.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Discuss the limitation, modification and deviations of Hammett equation. (7)

(b) Account for the following, compound I is more acidic than the compound II. (3)

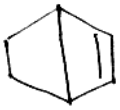


I II

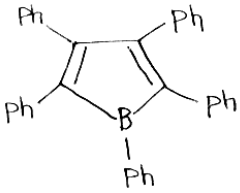
17. (a) Explain chan –ingold-prelog rules for assigning the configuration of a chiral carbon. (6)

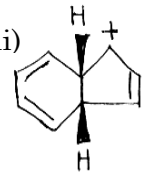
(b) Write a note on cram's rule. (4)

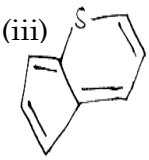
18. (a) Write the names for the following bicyclic compounds, (1.5+1.5+1)

(i) 
(ii)
(iii)

(b) Predict whether the following systems would be expected to show strong stabilization, weak stabilization by conjugation or destabilisation relative to localised modal structures. Explain the basis for your prediction. (6)

(i) 

(ii) 

(iii) 

19. (a) Explain why aniline is more reactive than acetanilide in electrophilic substitution. (2.5)

(b) Write short note on ortho/para ratio. (4)

(c) Give a detailed explanation for the following observation; Nucleophilic substitution in pyridine occurs readily at 2–and 4–position. (3.5)

20. (a) Discuss the following with mechanism, 'Acetolysis of anti –7- norbornenyl tosylate (I) is very much (10^{11} times) faster than that of its saturated analog (II).

(4)

(b) How can the E_1CB pathway be distinguished from the kinetically indistinguishable E_2 pathway? (2)

(c) Explain how nature of the substrate influence the relative reactivity in nucleophilic substitution reaction. (4)

AFS–0577

Sub. Code
7MCH1C2

M. Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Chemistry

INORGANIC CHEMISTRY —I

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the bond order of O_2?

2. Electron affinity of beryllium is almost zero. Why?

3. Calculate the number of bond pair and lone pair in $XeOF_4$.

4. Distinguish between sigma and pi bond.

5. Will Ag+ react more strongly with NH_3 and PH_3?

6. What is meant by levelling effect?

7. How is silane prepared?

8. The structure of boron nitride resembles that of graphite. But boron nitride acts as insulators Explain.

9. Explain the conductivity mechanism of p-type semi conductors.

10. What is line defect in solids?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) State and explain Fajan's rule.

Or

(b) Using Slater's rule, determine Z*for a 3p electron in P and 4s electron of Cu.

12. (a) Draw MO diagram of CO and calculate its bond order.

Or

(b) Using Hybridization theory, deduce the geometry of SF_4.

13. (a) Explain the factors affecting the strength of acid and bases.

Or

(b) Complete the following:

(i) K- $NH_3 \rightarrow$

(ii) HgI_2 +KI $\rightarrow$

(iii) BF_3 +HF$\rightarrow$

14. (a) Discuss the preparation and structure of P_4S_{10}.

Or

(b) What is inorganic benzene? How is it prepared? Mention any two properties.

15. (a) What is semiconductor? How are they classified? Mention its applications.

Or

(b) Write a note on Free electron theory of solids.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Use the Born-Haber cycle to calculate the enthalpy of formation of KBr, which crystallises in NaCl lattice. Use these data in the calculation: ΔH_{vap} (Br_2) = 29.8 KJ/ mol; Br_2 bond energy = 190.2 KJ / mol; and ΔH_{sub}(K) = 79 KJ / mol.

(b) What is Hydrogen bon? How are they classified? Explain with suitable examples. (5+5)

17. (a) Construct the wave function of Sp2 hybridisation.

(b) Explain the assumptions of VSEPR theory. How does this theory explain the variation of bond angle (5+5)

18. (a) Discuss the chemistry of Liquid HF as solvent.

(b) Explain HSAB principle. Discuss its applications (4+6)

19. Discuss the preparation and structure of P_4S_3 and S_4N_4.

20. (a) Write short notes on high temperature super conductors.

(b) Explain the role of semiconductors in solar energy conversion. (5+5)

AFS–0578

Sub. Code
7MCH1C3

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Chemistry

PHYSICAL CHEMISTRY - I

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define activity coefficient.

2. What is meant by Debye length?

3. What is fugacity? How does it relate to free energy?

4. Three moles of an ideal gas expand reversibly from a volume of 8 lit to 80 lit at a temperature at 300 K. Calculate the change in entropy.

5. What is triple point?

6. Define Le-Chatlier principle.

7. In what ways, M-B statistics differ from B-E statistics.

8. Define the term micro ensemble.

9. Define adsorption isobar and adsorption isostere.

10. What are protective colloids?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Derive Debye-Huckel- onsager equation.

Or

(b) What is overvoltage? Discuss the important applications of this phenomena.

12. (a) Derive Gibbs-Helmholtz equation for a process at constant pressure and at constant volume.

Or

(b) Define activity of a substance. The activity of 3.5 moles of a substance changes from 1.5 to 2.35.What would be the change in its free energy at 27°C.

13. (a) Draw and discuss the phase diagram of KI-H_2O system.

Or

(b) Derive the van't Hoff equation.

14. (a) Derive Boltzmann distribution law.

Or

(b) How are thermodynamic properties determined from partition function?

15. (a) Write a note on Gibbs adsorption isotherm.

Or

(b) What is Chemisorption? How does it differ form physisorption?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Derive the Butler-Volmer equation in the kinetics of the electrode reactions.

(b) Calculate the activity coefficients and mean ionic activity coefficients of Cu^{2+} and NO_3^- ions in 2.0 x10^{-3} M aqueous solution of $Cu(NO_3)_2$ (6+4)

17. (a) Derive Gibbs-Duhem equation

(b) Derive an expression for the entropy change accompanying the isothermal expansion of an ideal gas. (5+5)

18. (a) What are the factors affecting equilibrium constant?

(b) State the phase rule and derive Gibbs phase rule.

(5+5)

19. Discuss how the symmetry of wave functions of identical particles leads to the three kinds of quantum statistics-MB, BE and FD.

20. (a) Write a note on electro-osmosis

(b) Discuss briefly Langmuir theory of multilayer adsorption. (4+6)

AFS–0579

Sub. Code
7MCH1E1

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Chemistry

ELECTIVE: APPLIED CHEMISTRY

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define hardness of water.

2. What do you mean by Calgon conditioning?

3. Give the composition of LPG.

4. What is Cetane rating of Diesel?

5. Give the uses of neutron absorbing glass.

6. What are triple superphosphates?

7. What is molasses?

8. Write the synthesis of DDT.

9. Give the requisites of good lubricant.

10. What is electrodeless platting?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the sedimentation and coagulation treatment for domestic water supply.

Or

(b) Write short notes on reverse osmosis process in demineralization.

12. (a) Write notes on synthetic petrol.

Or

(b) Discuss the composition and uses of coal gas.

13. (a) Give a brief account of fertilizer industries in India.

Or

(b) Explain the composition and uses of optical glass and coloured glass.

14. (a) Describe the mode of action of gammexane.

Or

(b) Write short notes on manufacture of cane sugar.

15. (a) Explain the powder coating and antifouling coating method.

Or

(b) Discuss the fluid lubrication and extreme pressure lubrication.

Part C (3 × 10 = 30)

Answer any **Three** questions,

16. Discuss in derail on the various boiler troubles.

17. Describe (a). Refining of gasoline, (b). Octane rating of petrol and (c). Cetane rating of Diesel. (3+3+4)

18. Write notes on manufacturing of cement.

19. (a). Explain sugar estimation process. (5+5)

    (b). Give the mode of action of parathion and malathion.

20. (a) Write notes on graphite and molybdenum based solid lubricants.

    (b) Discuss the mechanism of drying of paint. (5+5)

AFS–0580

Sub. Code
7MCH1E2

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Chemistry

Elective: POLYMER CHEMISTRY

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Distinguish between organic and inorganic polymers

2. What is cross linked polymer? How is it formed?

3. Define repeating unit. Write the repeating unit of polychloroprene.

4. Mention the preparation of epoxy resin.

5. Give the isotactic configuration of a polymer

6. Mention any two processing properties of a polymer.

7. Mention the types of polymer degradation.

8. What is the effect of photo stabilizer on degradation of a polymer?

9. How are fibres classified in the consumer point of view?

10. What is the use of thermoforming process?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain Graft Polymer with suitable examples.

Or

(b) Discuss the mechanism of free radical polymerization.

12. (a) Give the structure of urea and write the preparation of urea formaldehyde.

Or

(b) How is PMMA prepared? Mention their uses.

13. (a) Write a note on heat distortion temperature.

Or

(b) What is meant by glass transition temperature? Give their relationship to melting point.

14. (a) Explain the role of antioxidants used in oxidative degradation of polymer.

Or

(b) Write short notes on ultrasonic and thermal degradation.

15. (a) Explain the role of plasticizer and fire retardants in plastic processing?

Or

(b) Distinguish between Thermosetting and thermoplastics.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. (a) Discuss the mechanism of group transfer polymerization.

    (b) Distinguish between organic and inorganic polymers. (5+5)

17. Describe the preparation and uses of the following.
    (a) Polyimides
    (b) polycarbonate
    (c) Kevlar (4+3+3)

18. (a) Derive an expression for the kinetics of free radical polymerization.

    (b) Describe the sedimentation method of determining molar mass of macromolecules. (5+5)

19. Write short notes on following polymerization techniques:
    (a) Bulk (b) Melt condensation. (5+5)

20. Using schematic diagram, explain the process of
    (a) Calendaring and (b) Film extrusion. (5+5)

AFS–0581

Sub. Code
7MCH2C1

M.Sc DEGREE EXAMINATION, APRIL 2018

Second Semester

Chemistry

ORGANIC CHEMISTRY – II

(CBCS – 2017 onwards)

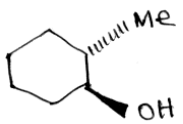
Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

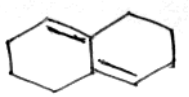
Answer **all** questions.

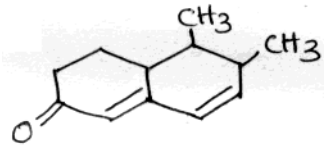
1. Bring out the difference between conformation and configuration.

2. Draw the clear conformational drawings for the following molecule, labelling each substituent as axial or equatorial.



3. Calculate the λ_{max} for the following molecules,

(a) 

(b) 

4. The carbonyl IR stretching frequency of cyclohexanone is 1715 cm^{-1} where as that of cyclopentanone is 1745 cm^{-1}. Explain.

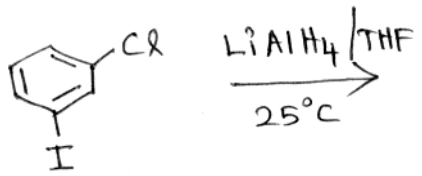
5. With example explain equivalent and non-equivalent protons.

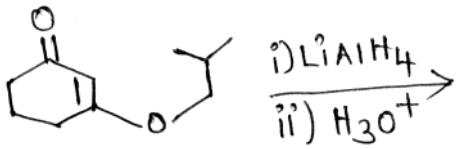
6. Identify the different types of protons in the following compounds,

   (a) $CH_3 - CBr_2 - CH_3$

   (b) $CH_3 - CH_2 - CH_2Cl$.

7. Write down the products for the following :

   (a) 

   (b) 

8. Explain how 1, 3-dithiane behaves as umpolung reagent with an example.

9. How cholestrol double and bond position is established?

10. Ergocalciferol does not give Diel's hydrocarbon when distilled with selenium, comment.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Discuss the stereochemistry and conformational features of 1,3-disubstituted cyclohexanes.

Or

(b) Discuss the conformational analysis of ethane with the help of a potential energy.

12. (a) Explain the factors influencing group frequencies in IR spectroscopy.

Or

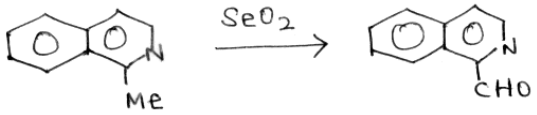
(b) Give an account on base peak, parent ion and metastable ion.

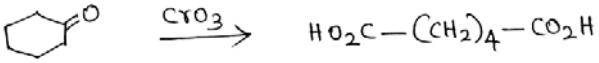
13. (a) Discuss the off-resonance decoupled and proton decoupled ^{13}C NMR spectra for 2-methyl butane. How does the proton-coupled signal for tetramethylsilane appear?

Or

(b) A compound with molecular formula $C_9H_{11}Br$ gives the following signals in the NMR spectrum; singlet = δ 7.25 (5H), doublet = δ 2,75(2H), multiplet = δ 3.40 (1H), doublet = δ 1.45(3H). Deduce the structure with reasoning.

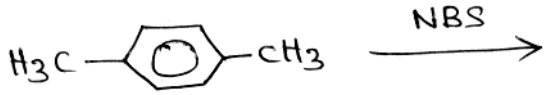
14. (a) Give the mechanism for the following conversion. (2.5 + 2.5)

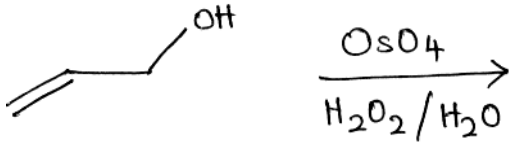
(i) 

(ii) 

Or

(b) Predict the product and give their mechanism for the following reaction : (2.5 + 2.5)

(i) 

(ii) 

15. (a) Discuss the synthesis of progesterone from ergosterol.

Or

(b) Indicate how the position of angular methyl groups in cholestrol established.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Write down the conformational enantiomers of 2, 3-dimethyl butane and 2, 2, 3-trimethyl butane. (3)

(b) Write a note on Eliel-Ro equation. (5)

(c) 1,1-disubstituted cyclohexanes can not exhibit any configurational isomerism. Explain. (2)

17. (a) Two isomers (I) and (II) having the molecular formula C_3H_6O exhibit the following peaks in the IR spectrum, (I) 1710 cm^{-1}, (II) 3300 cm^{-1}, 1640 cm^{-1}. Write the structures of I and II. (3)

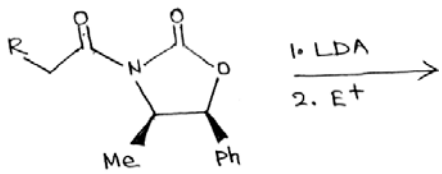
(b) Suggest molecular formulae for the fragment ions obtained from isobutane m/e 43, 28, 27? (3)

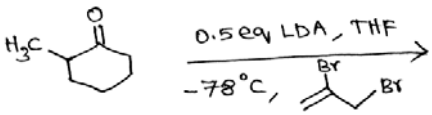
(c) Write notes on Retro-Diels-Alder reaction. (4)

18. (a) Discuss briefly on the influence of stereochemical factors on chemical shift of protons. (6)

(b) Write notes on spin-spin coupling. (4)

19. (a) Predict the product(s) (major and minor wherever applicable), with stereochemistry. (2.5 + 2.5)

(i) 

(ii) 

(b) Discuss the mechanism and application of Meerwein-Pondof-Verley reduction. (5)

20. (a) Write down the synthesis of ergocalciferol. (3)

(b) Describe the conversion of cholestrol into testosterone. (5)

(c) Draw the structure of an drosterone with proper stereochemistry. (2)

AFS–0582

Sub. Code
7MCH2C2

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Chemistry

INORGANIC CHEMISTRY – II

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. $[Cu(en)_2]^{2+}$ is more stable than $[Cu(NH_3)_4]^{2+}$. Why?

2. Write the formula of the following :

   (a) tris(ethylenediamine)rhodium(III) ion

   (b) *trans*-diamminetetraisothiocyantochromate(III) ion.

3. What is meant by inert complexes?

4. How does the electronic configuration of metal ion affect the rate of substitution reaction?

5. How does pH affect the binding of oxygen to hemoglobin?

6. Give any one example and its applications of blue copper proteins.

7. What is meant by reductive elimination reaction?

8. What are the conditions for a metal to act as catalyst?

9. Why is $^{238}U_{92}$ not suitable for nuclear fission reaction?

10. What are magic numbers?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) What is stepwise stability constant? How are they related to overall stability constant?

Or

(b) An octahedral complex of Fe^{2+} is diamagnetic. Is it an inner orbital or outer orbital octahedral complex?

12. (a) Discuss in detail the mechanism involved in the acid hydrolysis reaction.

Or

(b) Explain inner and outer sphere electron transfer reactions with suitable examples.

13. (a) Write an essay on *in-vivo* and *in-vitro* nitrogen fixation.

Or

(b) Discuss the structure of chlorophyll.

14. (a) Discuss the mechanism of Oxo's process.

Or

(b) Write short notes on ligand association and dissociation reactions.

15. (a) The activity of radioactive isotope falls to 12.5% in 90 days. Calculate the half life and decay constant.

Or

(b) Describe the applications of radioactive isotopes.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Discuss the postulates of VB theory. Bring out the defect of VB theory.

(b) State and explain Jahn-Telller theorem. What are its consequences? (5 + 5)

17. (a) What is trans effect? Explain the π bonding theory of trans effect.

(b) Discuss the mechanism of binuclear nucleophilic substitution reaction. (5 + 5)

18. (a) Explain the structure of ferridoxin. In what ways, it differs from rubredoxin?

(b) Compare the structure of hemoglobin and myolgobin. (5 + 5)

19. (a) What is Wilkinson catalyst? Bring out the importance of Wilkinson catalyst.

(b) What is Ziegler-Natta catalyst? How does it catalyze the polymerization reaction? (5 + 5)

20. (a) How are nuclear reactors classified? Explain the working of Breeder reactor.

(b) Explain the origin of stellar energy. (6 + 4)

AFS–0583

Sub. Code
7MCH2C3

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Chemistry

PHYSICAL CHEMISTRY – II

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What is the unit of rate constant of zero order reaction?

2. The rate constant for a first order reaction is $1.54 \times 10^{-3}\ s^{-1}$,Calculate its half life time.

3. CH_3Cl shows microwave rotational spectrum whereas CH_4 does not. Why?

4. How many normal modes of vibration are possible for the following molecules: SO_2 and CH_4?

5. Define mutual exclusion principle.

6. What are stokes lines?

7. Predict the ESR spectrum of benzene radical.

8. What is meant by spin decoupling?

9. Distinguish between symmetry elements and operation.

10. Explain why a set of numbers cannot form a group by a process of division.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Distinguish between order and molecularity of a reaction.

Or

(b) Discuss in detail the RRKM theory of unimolecular reactions.

12. (a) Discuss the isotope effect in rotational spectra.

Or

(b) Explain the concept of group frequencies.

13. (a) Explain the theory of Raman scattering.

Or

(b) Discuss the principle of PES and Draw the PES of Nitrogen molecule.

14. (a) Explain the role of lanthanide shift reagents in NMR spectroscopy.

Or

(b) State and explain Hyper fine splitting in ESR spectroscopy.

15. (a) Using group orthogonality theorem, derive the character table for C_{2v} point group.

Or

(b) Explain the various conditions must be satisfied to form a group.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. (a) Explain the terms temperature coefficient and energy of activation.

    (b) Discuss the Lindemann theory of unimolecular theory. (5+5)

17. (a) Sketch the normal modes of vibration of (i) H_2O (ii) CO_2 and determine which are IR –active/inactive and why?

    (b) Explain the instrumentation of microwave spectrometer. (6+4)

18. (a) Write a note on Laser Raman spectra.

    (b) State and illustrate with suitable potential energy curves the Frank Condon principle in the vibration spectrum of a diatomic molecule. Discuss briefly.

    (5+5)

19. (a) Discuss the principle and applications of NQR spectroscopy.

    (b) Explain the splitting pattern in Naphthyl radicals.

    (4)

20. (a) Establish the symmetry species of the normal modes of vibration NH_3 molecule.

    (b) Distinguish between reducible and irreducible representations. (5+5)

AFS–0584

Sub. Code
7MCH2E1

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Chemistry

Elective – ENVIRONMENTAL CHEMISTRY

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define: Smog.

2. What is acid rain?

3. What is meant by defluoridation?

4. Give the chemical composition of water bodies.

5. How soil is formed?

6. Define : Soil fertility.

7. What is the cause of Minamata disaster?

8. Write the effect of non-ionizing radiation on life.

9. What is an earthquake?

10. How do we prevent disaster?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the hydrological cycle of the environment.

Or

(b) Write short notes on the major sources of air pollution.

12. (a) Give the characteristics of various water resources.

Or

(b) Discuss the effect of organic-inorganic materials in water pollution.

13. (a) Give a brief account of nitrogen and pH analysis in soil.

Or

(b) Explain the composition and classification of soil.

14. (a) Describe the industrial pollution caused by fertilizer and chemical industries.

Or

(b) Write short notes on biochemical effects of cadmium and lead.

15. (a) Explain the types of cyclones. How cyclones are formed?

Or

(b) Discuss a case study of nuclear accident.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. (a) Describe ozone layer depletion.

    (b) Write short notes on the effect and control of air pollution. (5+5)

17. (a) Discuss the sampling and analysis of water pollution. (4)

    (b) Write notes on radioactive pollutants in water and its impact on aquatic environment. (6)

18. Write notes on soil pollution by

    (a) Fertilizers, (3+3+4)

    (b) Pesticides and

    (c) Heavy metal compounds.

19. (a) Give an account of Bhopal gas tragedy. (5+5)

    (b) Write notes on radiation protection and control.

20. (a) Write notes on human induced seismic activity. (4+6)

    (b) What is Tsunami? Why do it happen?

AFS–0585

Sub. Code
7MCH2E2

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Chemistry

Elective: COMPUTER IN CHEMISTRY

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the different types of main memory?

2. When do we use ROM?

3. What are the types of 'C' constants?

4. What are fundamental data types?

5. Give the syntax for assignment statement.

6. What is ternary operator?

7. What is web server?

8. What do you mean HTTP?

9. What is scope of cheminformatics?

10. Mention any two chemical resources used in cheminformatics.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the various generations of computers.

Or

(b) List a few commonly used input devices and explain them briefly.

12. (a) Write a C program to calculate pH of a solution.

Or

(b) Write a C program to evaluate cell constant.

13. (a) Using loop statement, Write a C program to calculate mean activity coefficient of an electrolytes.

Or

(b) Write a C program to calculate empirical formula of hydrocarbon.

14. (a) List out the procedures used in composting and sending of an e-mail.

Or

(b) Write a note on popular websites in chemistry.

15. (a) Write short note on computational chemistry.

Or

(b) Using ChemDraw software, list out the procedures to draw the simple two-dimensional representations of organic molecules.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a C program to calculate normality, molality and molarity of a given solution. (5+5)

17. Write a C program to calculate

    (a) lattice energy using Born-Lande equation and

    (b) first order rate constant for a given reaction. (5+5)

18. Write a C program to calculate different velocities of gases.

19. (a) List out the tips for effective use e-mail using smile keys

    (b) Write a note on different search engines (5+5)

20. (a) What are the applications of drug discovery in cheminformatics

    (b) Explain the major goals and strategies of drug discovery (5+5)